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Torque and Force sensor systems

User's guide

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1. Introduction

This document is a user's guide for the following items:

- The analog torque sensing system.
- The digital torque sensing system.

Each of the torque sensing systems is composed of a set of 2 force sensing sub-systems, supposed to be used simultaneously. The torque sensing systems may be interfaced to the FC HARP H1 board or to a simpler Arduino board.

2. Scope

This document covers the operation of both the analog and the digital version of the force sensing systems. This document does not cover the mechanical mounting and integration of the force sensors.

3. Precautions for use

Not complying with the following precautions may lead to the systems permanent destruction or degradation:

3.1. Electro-static discharges

The systems in this delivery are bare electronic boards suitable for integration by the customer. As such, sensitive electronic components are exposed to manipulation. Precautions shall be used to limit the risk of electro-static discharges on the electronic parts.

3.2. Mechanical constraints

The sensors provided in the delivery are sensitive to mechanical constraints (especially the digital sensor). The sensors shall be manipulated with the most extreme care and shall not be exposed to mechanical constraints exceeding their specifications (see respective datasheets).

3.3. Power supply

While the analog version can be powered with 3.3V to 5V, the digital sensor accepts only 3.3V. The same voltage standard shall be used for the sensors' power supply and the I/Os.

3.4. Connections

All connectors orientation shall be respected.

4. Analog torque sensing system

4.1. Generalities

The analog torque sensing system is composed of a set of 2 analog force sensing sub-systems. Each of the force sensing sub-systems is composed of:

- A load cell, reference TAL221, 100g nominal range.
- A HX711 load cell amplifier and digital interface IC break-out board.

The 2 HX711 boards are mounted on a single hosting circuit board. 2 load cells are connected to the hosting circuit board. The hosting circuit board is connected to the interface board (not shown on the picture), supplies for building a wiring harness for connection to the FC HARP H1 interface board are provided for this purpose.



The board can also be connected to an Arduino board thanks to some provided cables. An Arduino sketch is provided as part of the delivery so as to demonstrate the system's operation and ease its integration in the final setup.

4.2. Hosting circuit board description

The board allows for simultaneously reading from 2 distinct load cells. It is hosting 2 independent HX711 circuits: circuit 1 and circuit 2.

Each of the HX711 avails 2 channels for load cells connections: channel A and channel B. The following gain settings are available for the channels:

- Channel A: 64 or 128, selectable by software.
- Channel B: 32, fixed.

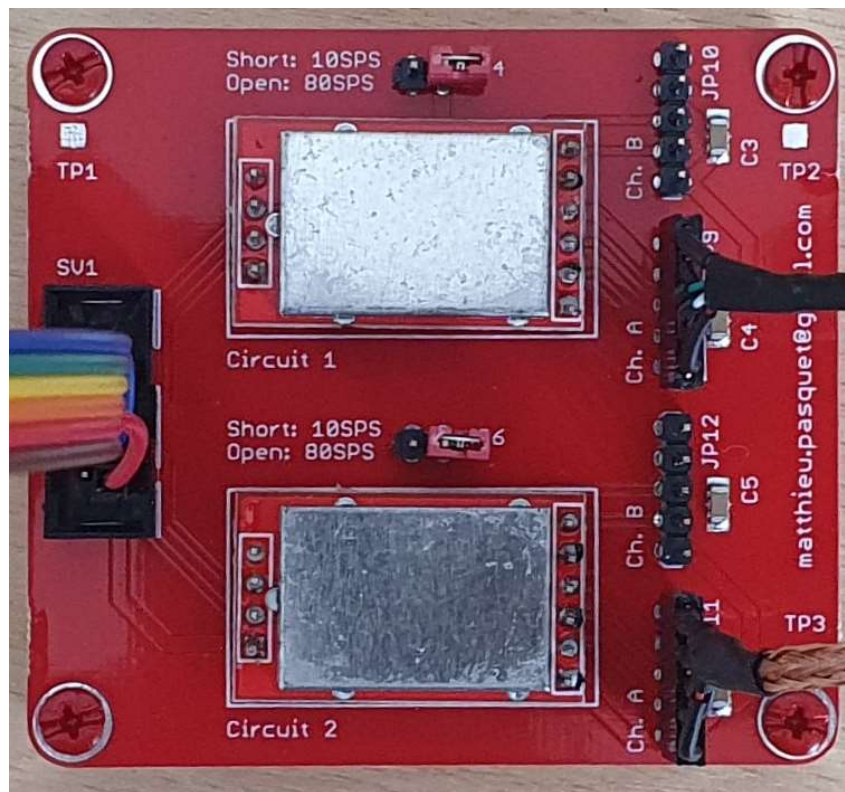
The load cells shall be connected to the suitable channel accordingly to the targeted gain.

The HX711 allows for selecting among 2 data rates:

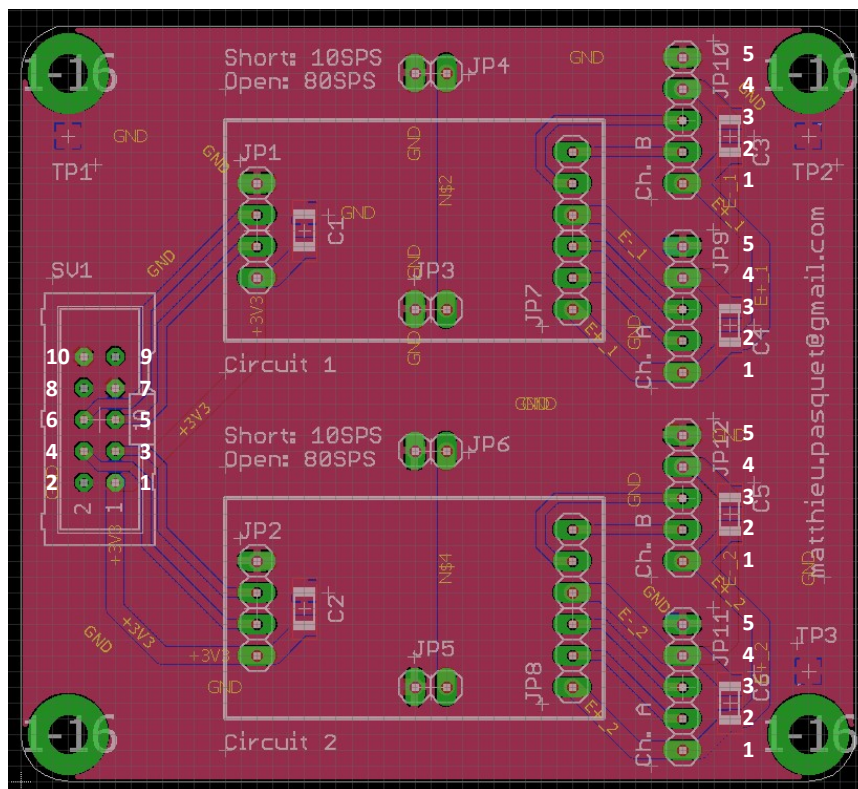
- 10 SPS.
- 80 SPS.

The data rate can be selected on the board using JP4 and JP6.

Circuit board:



The hosting circuit board avails the following connectors:



SV1: Interface connector to interface board (HARP or Arduino).

Pin	Function
1	Power Supply +3.3V or +5V
2	Not used
3	Circuit 2 Clock
4	Circuit 2 data
5	Circuit 1 Data
6	Circuit 1 Clock
7	GND
8	Not used
9	Not used
10	GND

JP9: Circuit 1, Channel A load cells connector.

JP10: Circuit 1, Channel B load cells connector.

JP11: Circuit 2, Channel A load cells connector.

JP12: Circuit 2, Channel B load cells connector.

JP9, JP10, JP11, JP12 share the same pinout:

Pin	Function
1	Excitation +
2	Signal -
3	Signal +
4	Excitation -
5	Shielding (GND)

Additionally, the board avails jumpers to select the data rate:

JP4: data rate selection for circuit 1.

JP6: data rate selection for circuit 2.

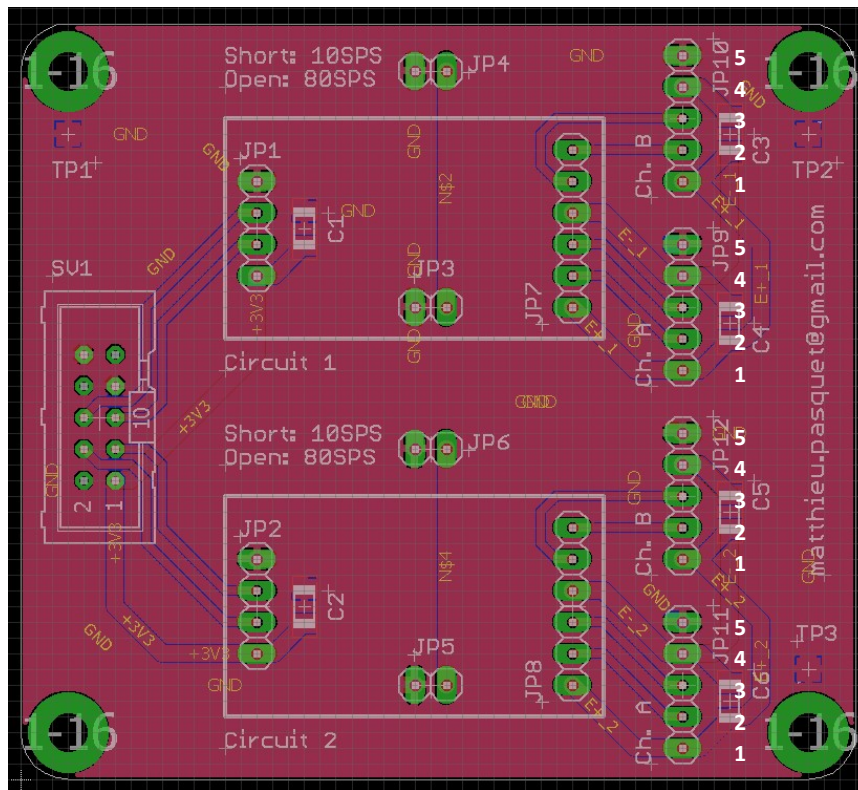
When left opened, the data rate is 80 Samples Per Second (SPS).

When shorted, the data rate is 10 Samples Per Second (SPS).

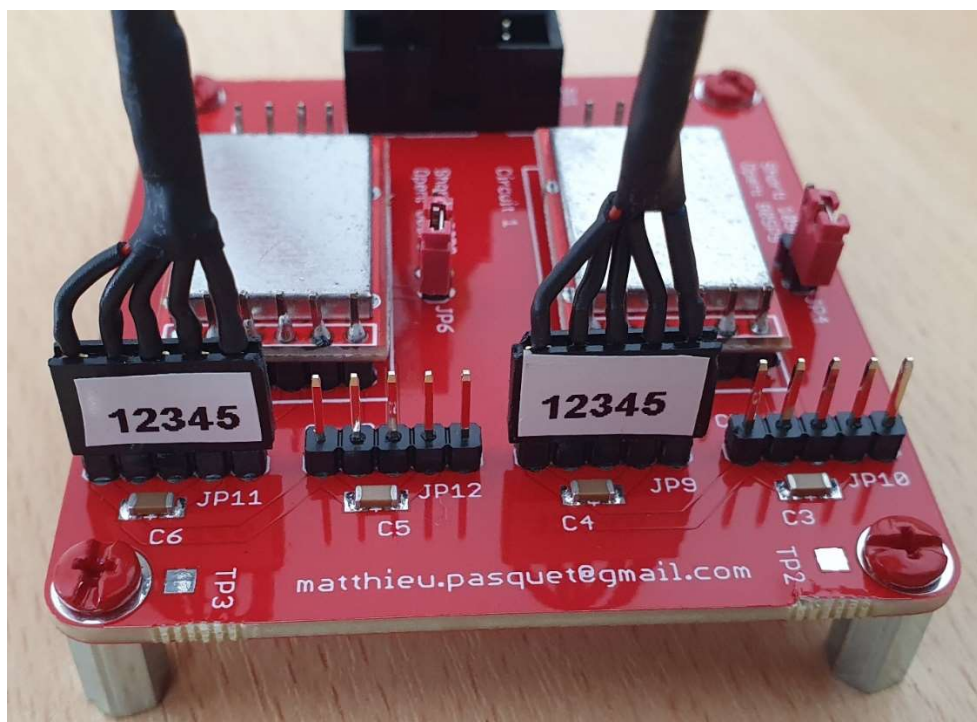
Overall, the lower the gain and the lower the data rate, the lower the reading noise.

4.3. Load cells connection

The load cells connectors' pin numbers are labelled on the load cell connector. The connector orientation shall be respected:

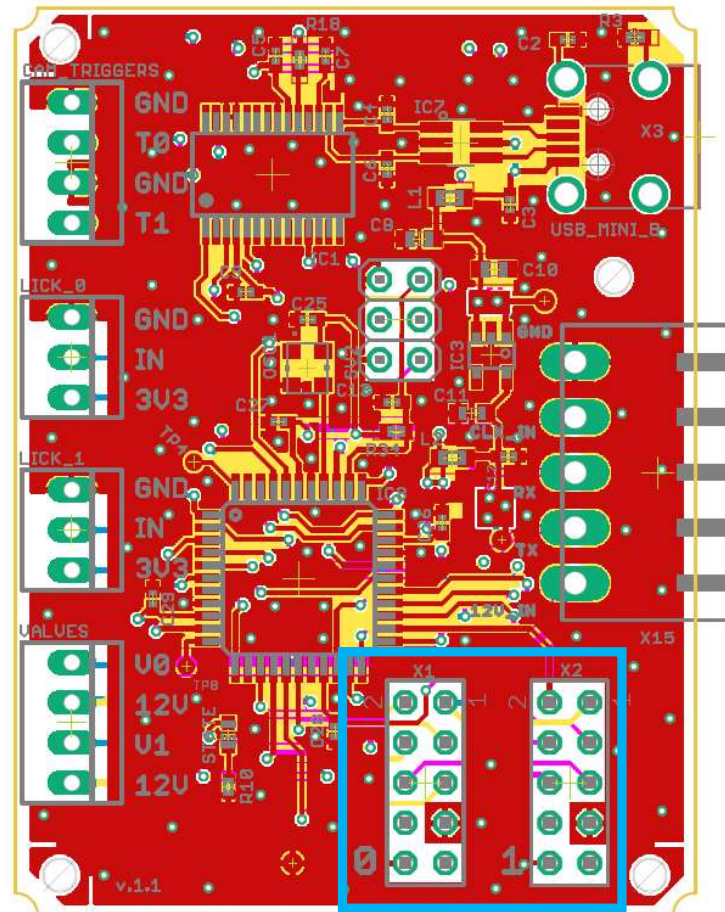


Please see the following example for channel A connection on both circuit 1 and circuit 2:



4.4. Connection to the HARP H1 interface board

The sensor circuit board may be connected to the FC HARP H1 board (HARP H1 firmware update required). Wiring supplies (cable and connectors) are provided for building a harness for this purpose. The sensor board may be connected to either the X1 or X2 connectors on the HARP H1 board (blue square):



The wire harness shall implement the following connectivity:

Sensor Board Pin	Sensor Board Function	HARP H1 Pin
1	Power Supply +3.3V or +5V	1
2	Not used	2
3	Circuit 2 Clock	3
4	Circuit 2 data	4
5	Circuit 1 Data	5
6	Circuit 1 Clock	6
7	GND	7
8	Not used	8
9	Not used	9
10	GND	10

4.5. Testing with the Arduino sketch

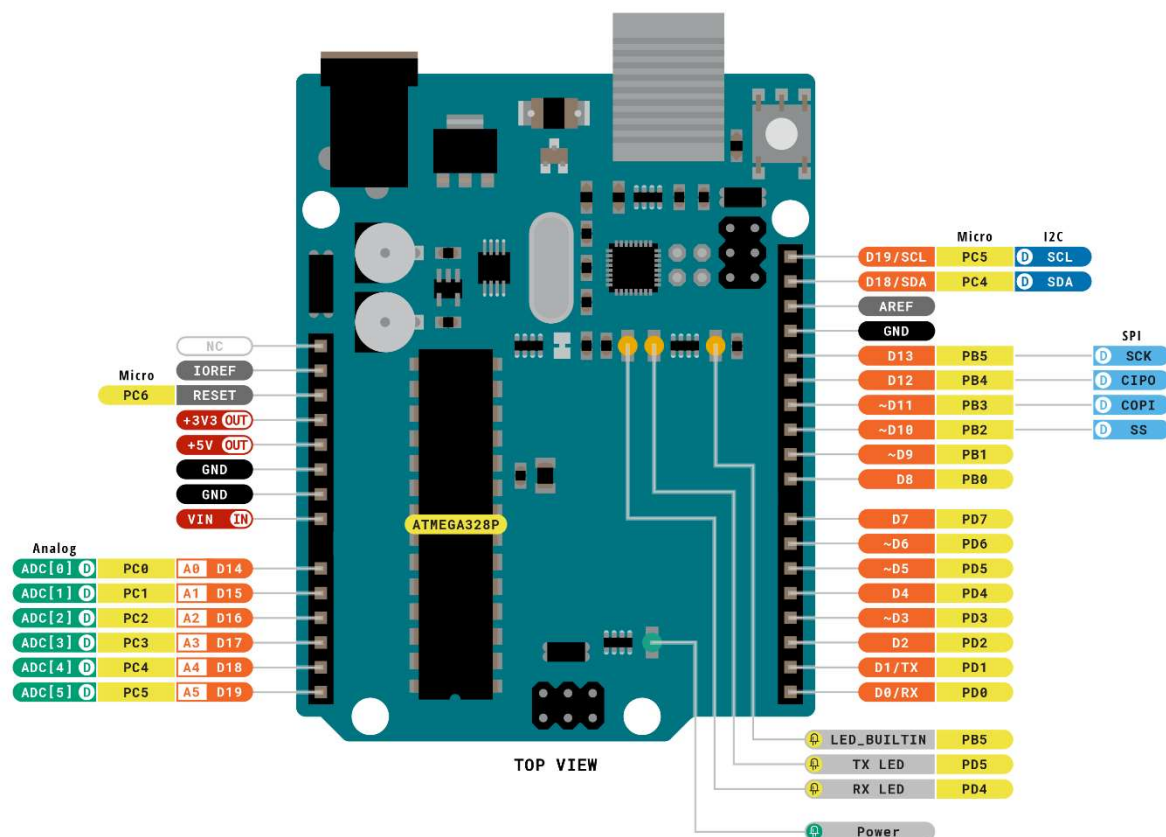
An example Arduino sketch is provided for testing purposes.

4.5.1. Wiring to the Arduino UNO R3 board

Wires connection is performed with the Arduino board not powered nor connected to the USB.

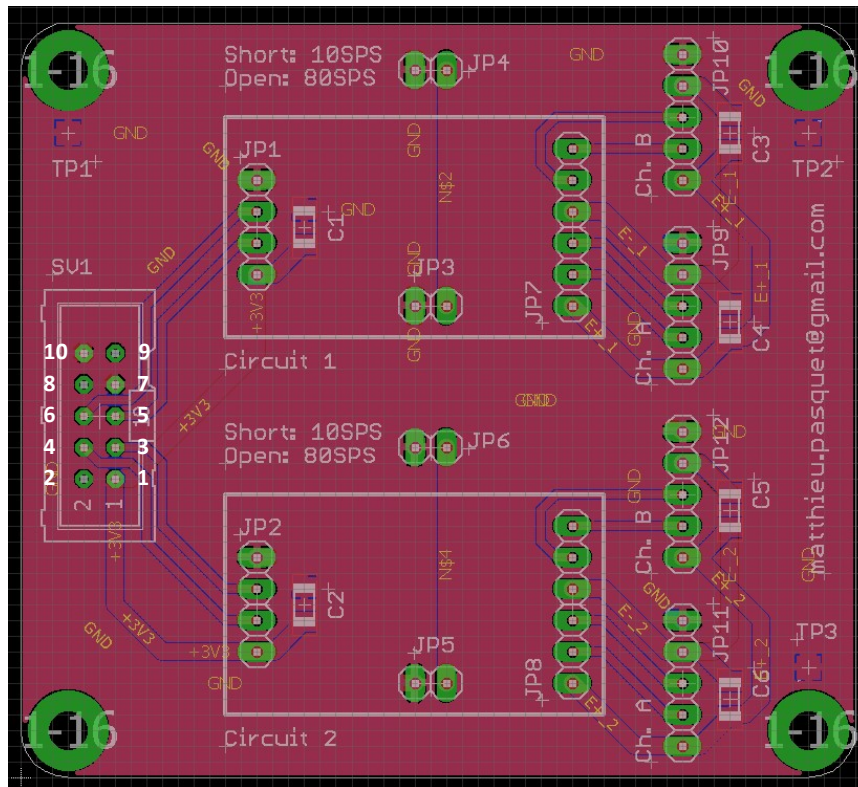
4.5.2. Introduction

The testing relies on the use of an Arduino R3 board. In the example, a single Arduino board is managing 2 HX711 ICs, each connected to a single TAL221 100g range load cell. The analog interface board accepts a power supply of 3.3V or 5V, in this example the board will be powered with 5V, from the Arduino.



Legend:	Digital	I2C
Power	Analog	SPI
Ground	Main Part	Analog


 ARDUINO
 ARDUINO UNO REV3
 SKU code: A000066
 Pinout
 Last update: 6 Oct, 2022



The systems shall be connected as follows:

Analog interface board pin	Analog interface board pin number	Arduino Uno R3 Pin
+3.3V/+5V	1	+5V
GND	7 and 10	GND
Circuit 1 Clock	6	PD3 (3)
Circuit 1 Data	5	PD2 (2)
Circuit 2 Clock	3	PD5 (5)
Circuit 2 data	4	PD4 (4)

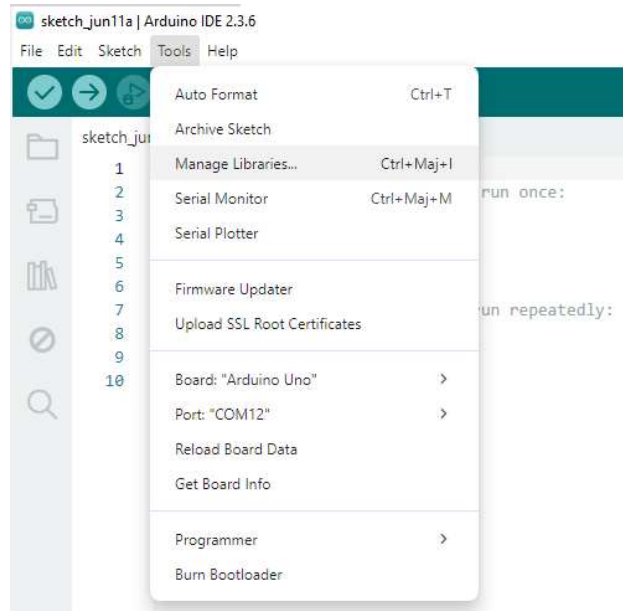
2 load cells shall be connected to the suitable channel on circuit 1 and circuit 2. The desired data rate shall be set using the JP4 and JP6 jumpers.

After all connections and settings are performed, the Arduino R3 board can be connected to a computer's free USB port.

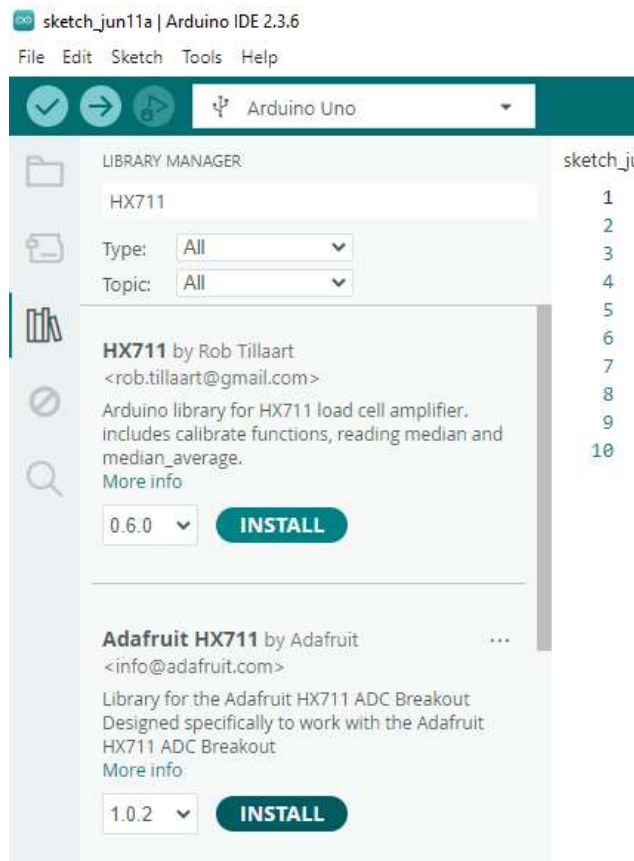
4.5.3. Arduino code running

The following procedure shall be followed:

- 1) The Arduino IDE shall be downloaded from <https://www.arduino.cc/en/software/> and installed.
- 2) Start the Arduino IDE and select Tools → Manage Libraries:



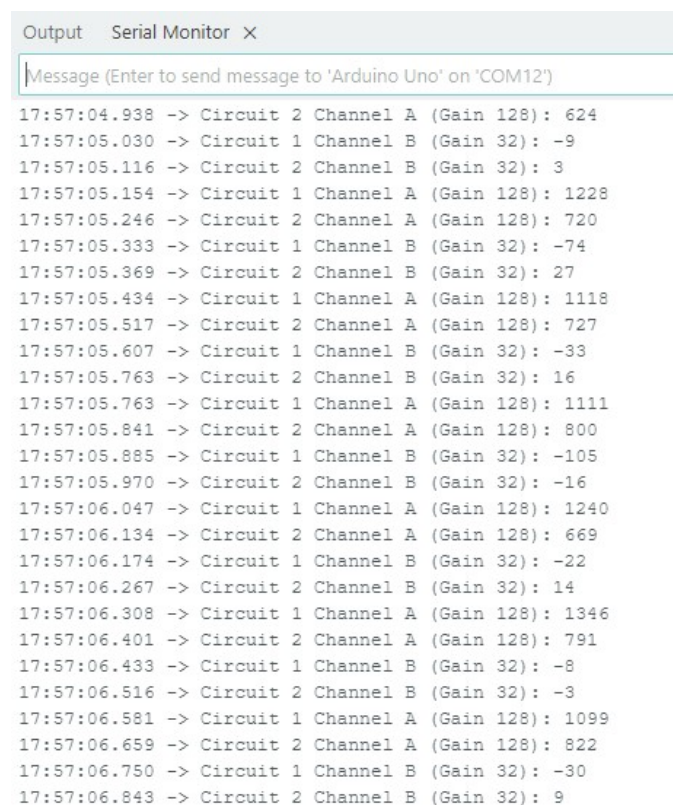
- 3) Search for “HX711” libraries and install the “Adafruit HX711” libraries.



- 4) Open the provided test sketch: File→Open, navigate to and open the provided "sketch_HX711.ino" sketch file.
- 5) Assuming the Arduino R3 board has been wired correctly to the HX711 board and it is connected to USB, select the suitable "Arduino Uno" board and click the arrow "upload" button:



- 6) When the upload is completed, start the serial monitor: Tools→Serial Monitor. The system will display the measurements:



4.5.4. HX711 library

The preferred installation method is by fetching the library from within the Arduino IDE. Alternatively, the library can be downloaded from https://github.com/adafruit/Adafruit_HX711.

5. Digital system

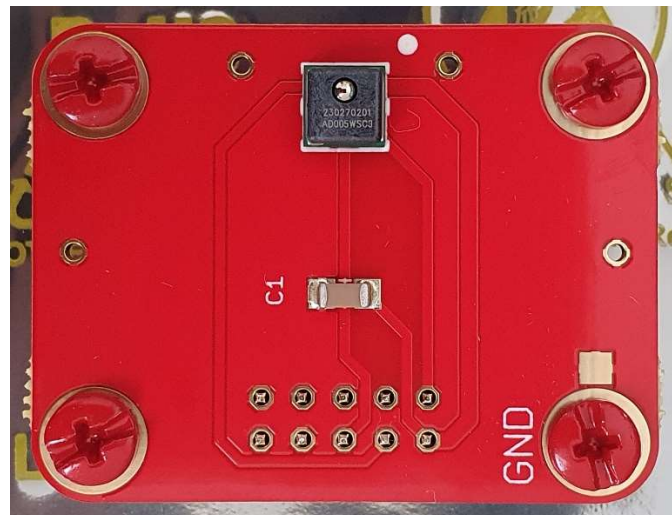
5.1. Generalities

The digital torque sensing system is composed of a set of 2 identical digital force sensing sub-systems. Each of the force sensing sub-systems is composed of:

- A single FMAMSDXX005WCSC3 3.3V 5N range SPI interface digital sensor.
- A connector to the interface board.
- All mounted on a circuit board.

The circuit board is connected to the interface board (FC HARP H1 or Arduino), supplies for building a wiring harness for connection to the FC HARP H1 interface board are provided for this purpose.

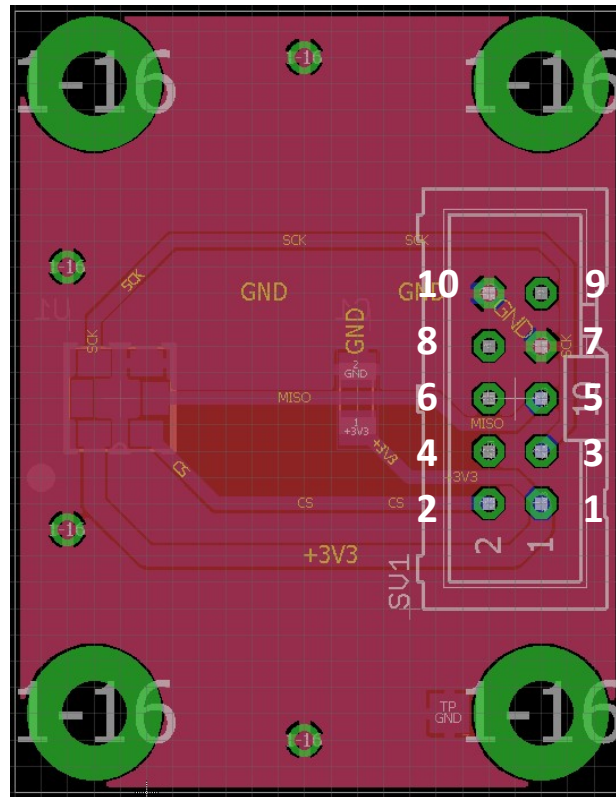
The board can also be interfaced to an Arduino board. The selected Arduino board **MUST** have **3.3V** IOs, and the sensor board **MUST** be powered with **3.3V**. An Arduino sketch is provided as part of the delivery so as to demonstrate the system's operation and ease its integration in the final setup.



5.2. Circuit board description

A single force sensor is mounted on each board.

The circuit board avails the following connector:

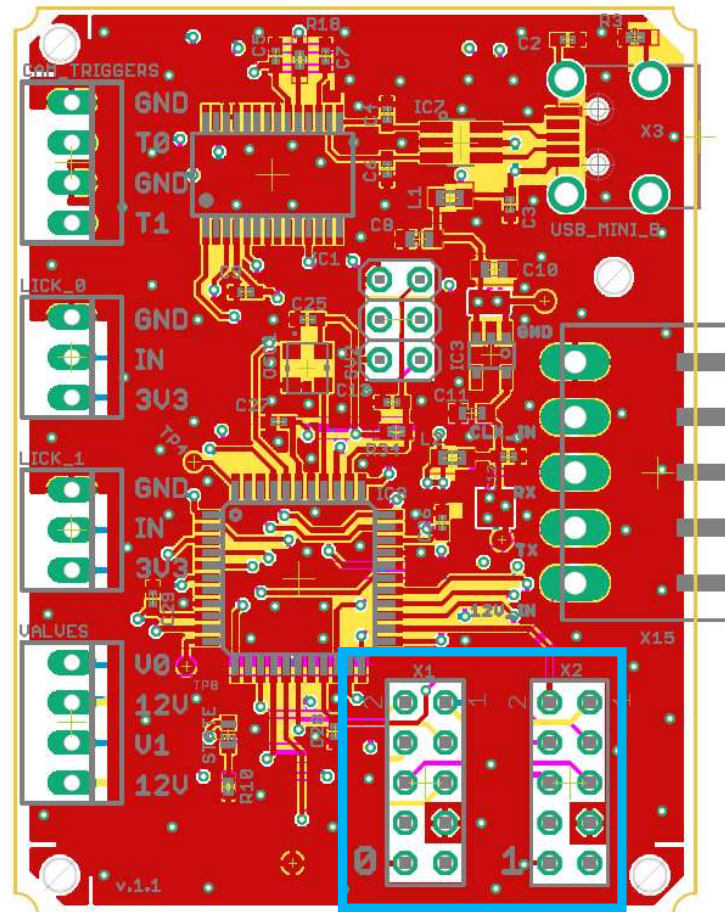


SV1: Interface connector to the interface board.

Pin	Function
1	Power Supply +3.3V
2	CS/SS
3	SCK
4	Not used
5	MISO
6	Not used
7	GND
8	Not used
9	Not used
10	GND

5.3. Connection to the interface board

Two sensor circuit boards may be connected to the FC HARP H1 board (HARP H1 firmware update required). Wiring supplies (cable and connectors) are provided for building a harness for this purpose. Each of the sensor boards may be connected to the X1 or X2 connectors on the HARP H1 board (blue square, one sensor per connector):



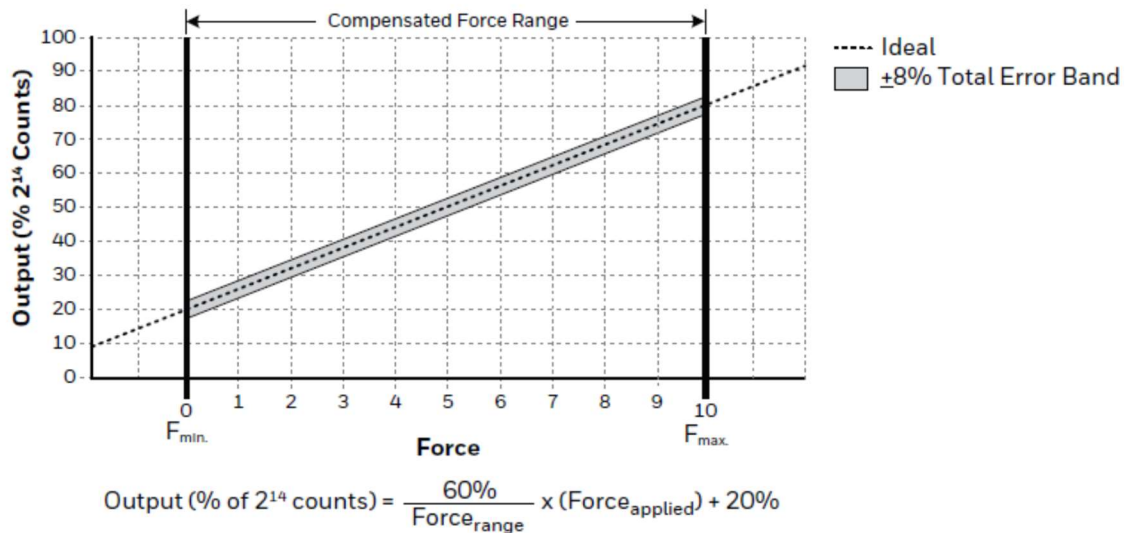
The wire harness shall implement the following connectivity:

Sensor Board Pin	Sensor Board Function	HARP H1 Pin
1	Power Supply +3.3V	1
2	CS/SS	2
3	SCK	3
4	Not used	4
5	MISO	5
6	Not used	6
7	GND	7
8	Not used	8
9	Not used	9
10	GND	10

5.4. Sensor output

The sensor datasheets provide the following transfer function:

FIGURE 1. TRANSFER FUNCTION LIMITS



The transfer function can be interpreted as follows: at 0N force input, the sensor digital output count is about 20% of the full range (the full range is 16384). Thus expect an output value around 3277 at 0N force.

Additionally, the following formula are provided:

Equation 1: Force Sensor Transfer Function

$$\text{Output} = \frac{(\text{Output}_{\text{max}} - \text{Output}_{\text{min}})}{\text{Rated Force Range}} \times (\text{Force}_{\text{applied}}) + \text{Output}_{\text{min}}$$

Rearranging this equation to solve for force provides Equation 2:

Equation 2: Force Output Function

$$\text{Force} = \frac{(\text{Output} - \text{Output}_{\text{min}})}{(\text{Output}_{\text{max}} - \text{Output}_{\text{min}})} \times \text{Rated Force Range}$$

Where:

Output_{max} = output at maximum force [counts]

Output_{min} = output at minimum force [counts]

Force_{rated} = maximum value of force range (N, lb, g, or kg)

Force_{applied} = force being measured by sensor (N, lb, g, or kg)

Output = digital force reading [counts]

Example: Calculate the force for a 10 N force sensor with a 10% to 90% calibration and a force output of 6880 (decimal) counts:

Output_{max} = 14745 counts (90% of 2^{14} counts or 0x3999)

Output_{min} = 1638 counts (10% of 2^{14} counts or 0x0666)

Force_{rated} = 10 N

Force = force in N

Output = 6880 counts

$$\text{Force} = \frac{[(6880 - 1638) \times 10]}{(14745 - 1638)}$$

$$\text{Force} = \frac{(5242 \times 10)}{13107}$$

Force = 4 N

This equation is implemented in the provided example Arduino sketch, adapted to the used sensor reference (option C: transfer function 20% to 80%).

The sensor datasheet specifies a “response time” of 0.84ms maximum. The sensor readout frequency should be limited to 1190Hz.

5.5. Testing with the Arduino sketch

An example Arduino sketch is provided for testing purposes.

5.5.1. Wiring to the Arduino ZERO board

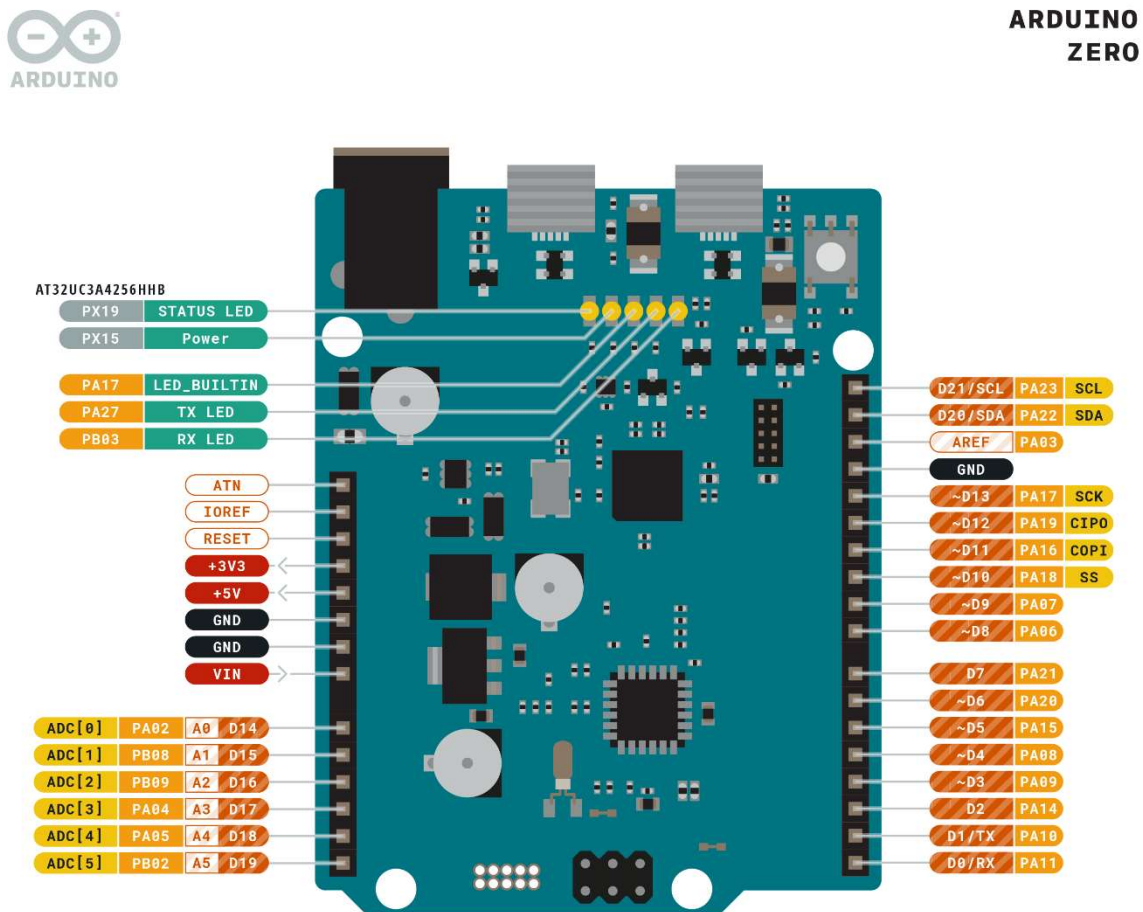
The digital sensor is a 3.3V system. It MUST be powered with 3.3V and the IOs voltage standard used to communicate with it MUST be 3.3V.

The Arduino ZERO board has 3.3V IOs and can provide 3.3V power. This board will be used in this example.

Wires connection is performed with the Arduino board not powered nor connected to the USB.

5.5.2. Introduction

The testing relies on the use of an Arduino ZERO board. In the example, a single Arduino board is managing one single force sensor board. The digital interface board requires a power supply of 3.3V, in this example the sensor board will be powered with 3.3V, from the Arduino.

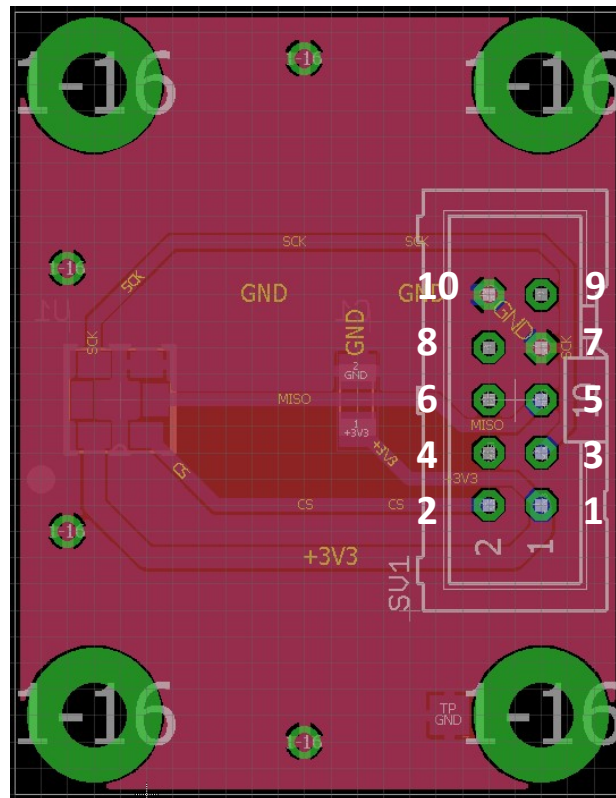
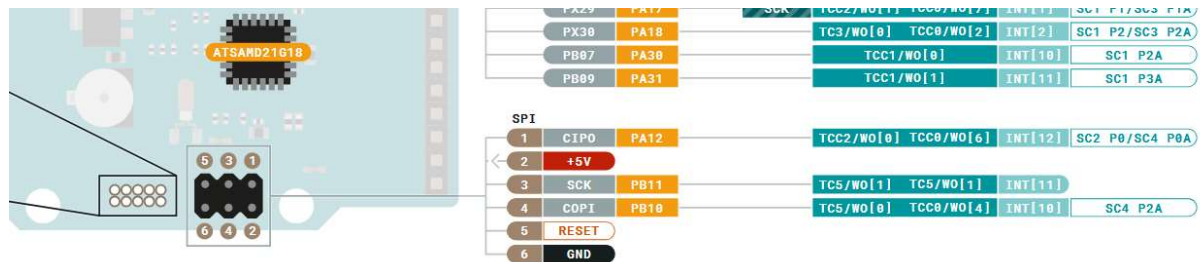


Ground	Internal Pin	Digital Pin	Microcontroller's Port
Power	SWD Pin	Analog Pin	
LED	Other Pin	Default	

ARDUINO.CC



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The systems shall be connected as follows:

Digital interface board pin	Digital interface board pin number	Arduino ZERO Pin
+3.3V	1	+3.3V
GND	7 and 10	GND
MISO	5	SPI connector 1
SCK	3	SPI connector 3
CS/SS	2	DO

After all connections and settings are performed, the Arduino R3 board can be connected to a computer's free USB port.

5.5.3. Arduino code running

The following procedure shall be followed:

- 1) The Arduino IDE shall be downloaded from <https://www.arduino.cc/en/software/> and installed.
- 2) Start the Arduino IDE and open the provided test sketch: File→Open, navigate to and open the provided “sketch_Digital.ino” sketch file.
- 3) Assuming the Arduino ZERO board has been wired correctly to the sensor board and it is connected to USB, select the suitable “Arduino ZERO (Native USB)” board and click the arrow “upload” button:



- 4) When the upload is completed, start the serial monitor: Tools→Serial Monitor. The system will display the measurements:

```
10:17:29.679 -> Raw Temperature (digital count): 847
10:17:29.679 -> Calibrated Temperature (°C): 32.76
10:17:29.679 ->
10:17:30.181 -> status (0: Normal operation, valid data): 0
10:17:30.181 -> Raw Force (digital count): 4881
10:17:30.181 -> Calibrated Force (N): 0.82
10:17:30.181 -> Raw Temperature (digital count): 847
10:17:30.181 -> Calibrated Temperature (°C): 32.76
10:17:30.181 ->
10:17:30.695 -> status (0: Normal operation, valid data): 0
10:17:30.695 -> Raw Force (digital count): 4881
10:17:30.695 -> Calibrated Force (N): 0.82
10:17:30.695 -> Raw Temperature (digital count): 847
10:17:30.695 -> Calibrated Temperature (°C): 32.76
10:17:30.695 ->
```

6. Building the wiring harnesses

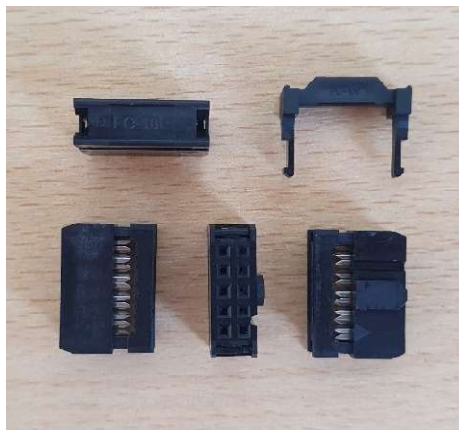
6.1. For the HARP H1 board

The following supplies are part of the delivery and can be used to build the wire harness to connect each sensor board to the HARP H1:

- 5m Flat 10 conductors ribbon cable.

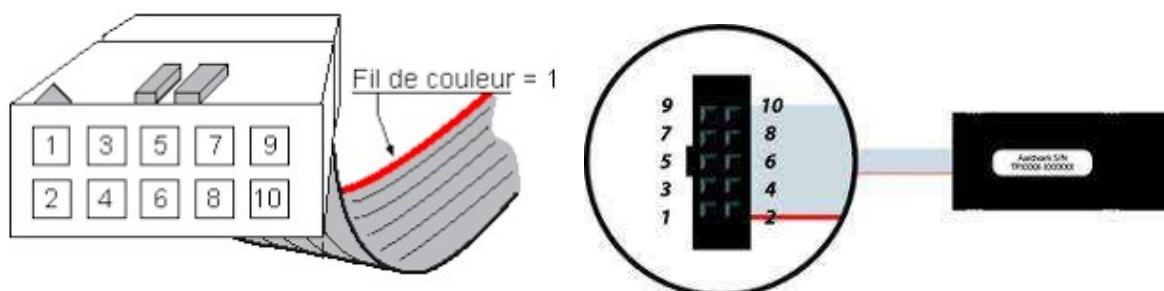


- 30 HE10 crimp-mount connectors with cable strain reliever.



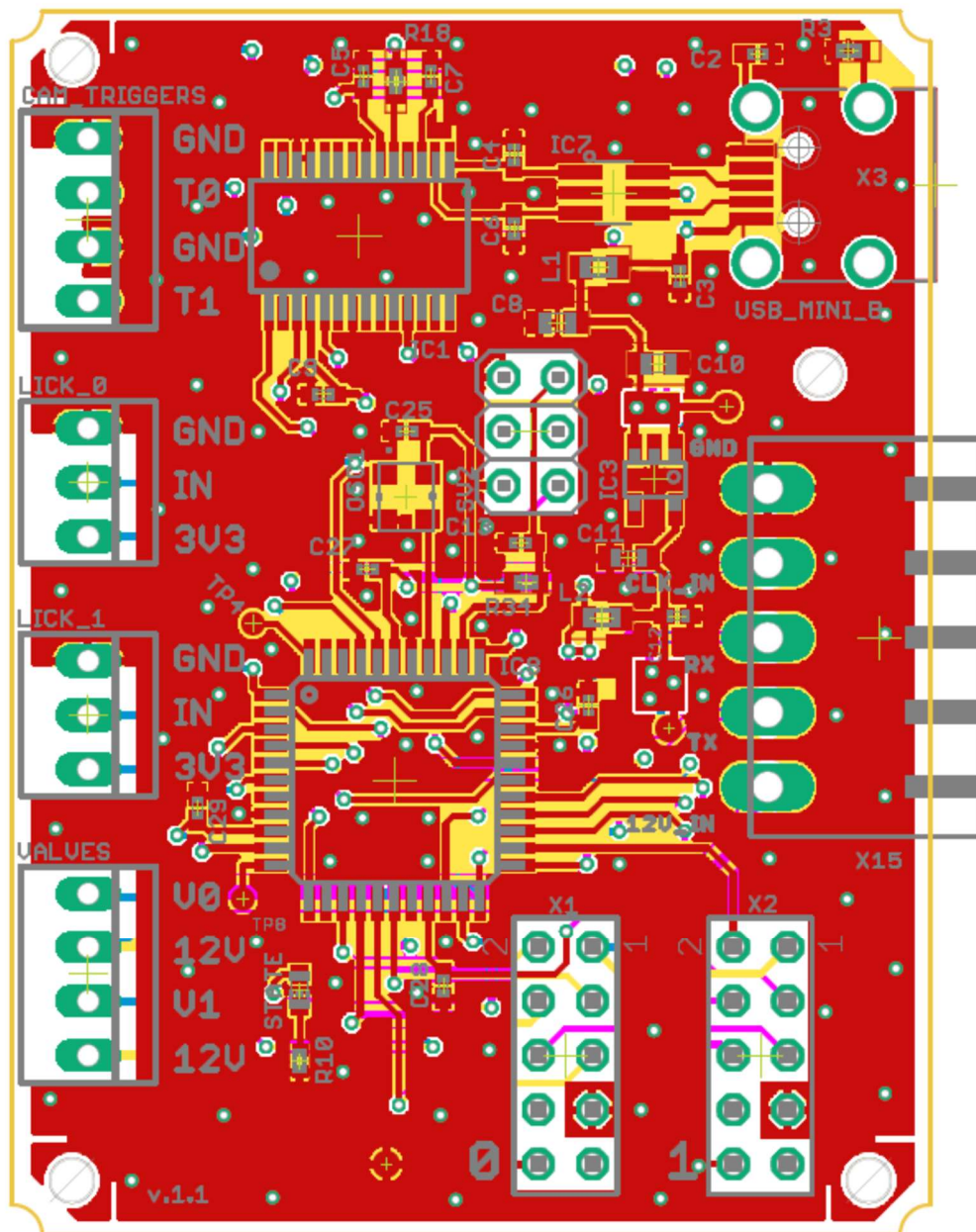
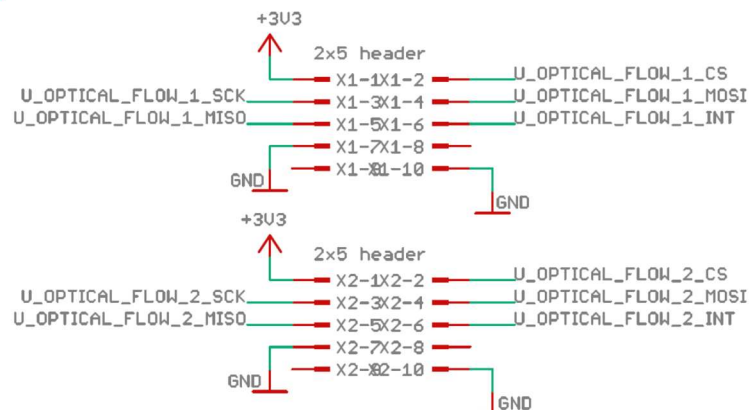
The following tutorial demonstrates how to crimp mount the connector to the cable (the specialized tool can be replaced by a small vise): <https://www.youtube.com/watch?v=4pDRrrpAcGQ>

The standard connector and cable pinout and orientation are the following:

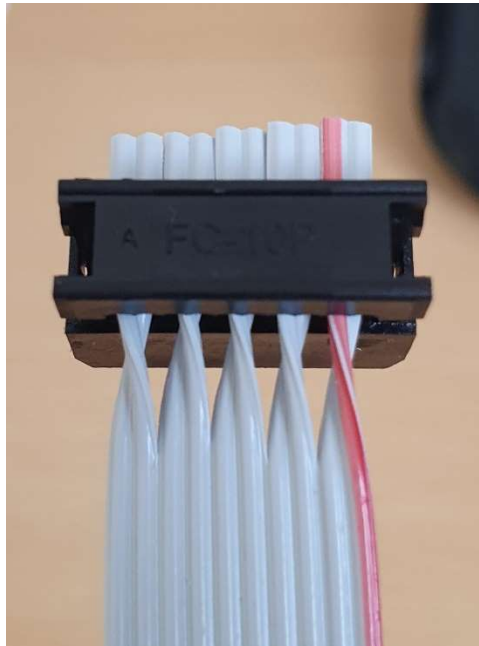


The HARP H1 board pinout is the following:

Optical Flows



Please notice that the way the X1 and X2 connectors are drawn on the HARP PCB (ie with pins 1 and 2 rows reversed) suggests these connectors are mounted on the bottom side of the circuit board. Shouldn't that be the case, ie connectors mounted on top, the wiring harness shall take this non-standard design in consideration and the cable shall be twisted as follows:



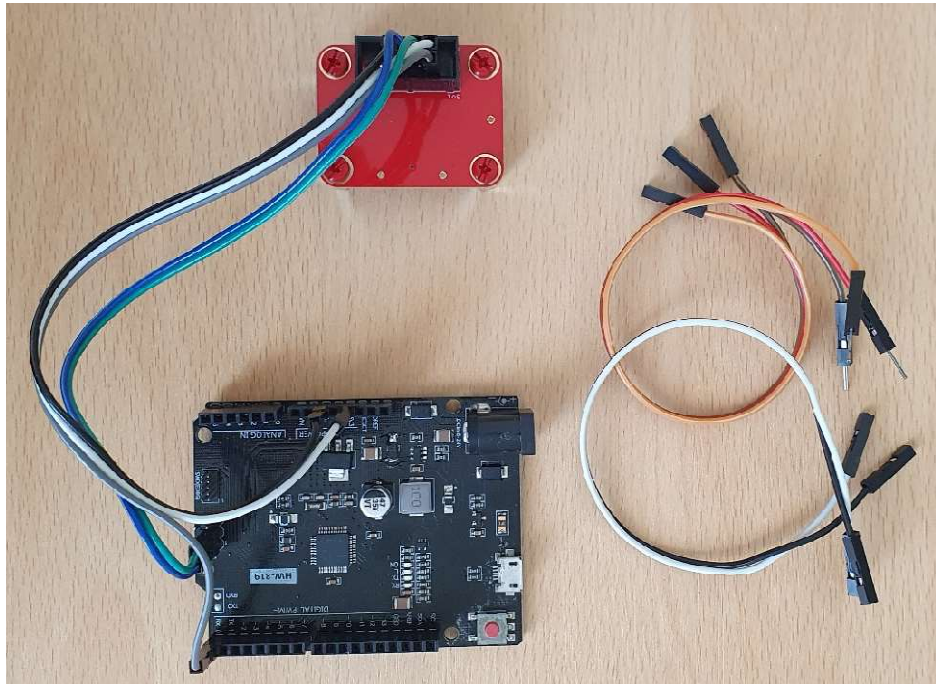
6.2. For the Arduino boards

A set of cables with male or female “Dupont” style pin connectors is provided for connecting the digital or analog boards to the appropriate Arduino board:



Follow the instructions in the relevant “wiring” chapter for proceeding with the connection.

Overall, the setup shall look like this (digital sensor on Arduino Zero board):



7. Bibliography

All the following documents are included in the delivery:

TAL221 100g load cell: TAL221.pdf

HX711 load cell amplifier and digital interface IC: hx711F_EN.pdf

FMAMSDXX005WCSC3 Sensor data sheet: hon-ia-hss-force-fma-series-datasheet-32347833-ciid-181799.pdf

FMAMSDXX005WCSC3 sensor SPI interface specification: sps-siot-force-spi-comms-tech-note-008290-ciid-156712.pdf

FMAMSDXX005WCSC3 sensor mechanical coupling: sps-siot-coupling-tech-note-008333-ciid-181807.pdf

FMAMSDXX005WCSC3 sensor over-force note: sps-siot-overforce-tech-note-008334-ciid-181806.pdf